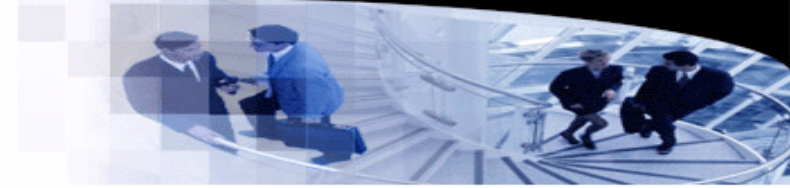




武汉大学
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Programming Languages and Compilers

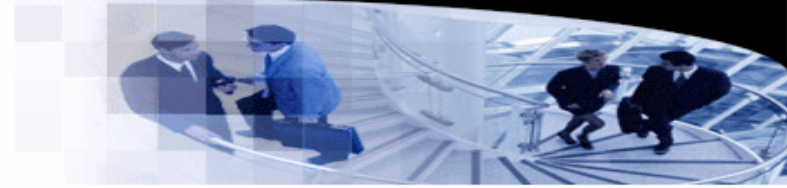
Lecture 24. Live Variable Analysis

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Live Variable Analysis

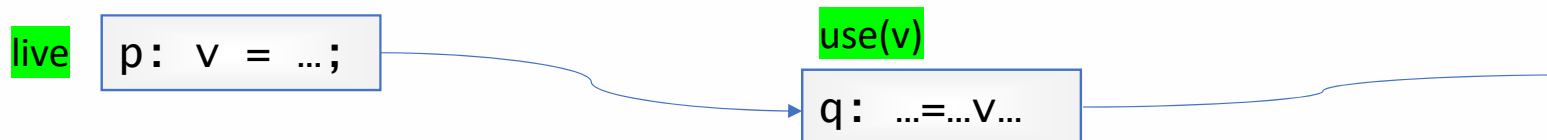


- **Liveness**

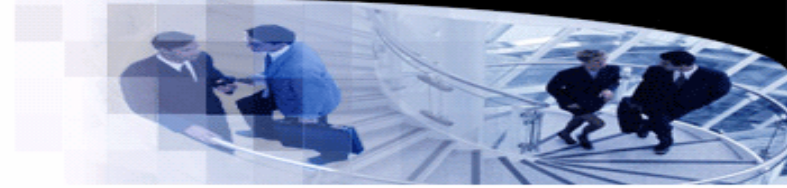
- A data-flow property of variables.
- Is the value of a variable needed?

- **Live variable analysis**

- Given a variable v at a program location p , live variable analysis answers whether it is alive.
- If it is alive, then there is a path starting from p , such that v is **used** somewhere on the path.
- If it is not alive, then v is never used on any path starting from p .



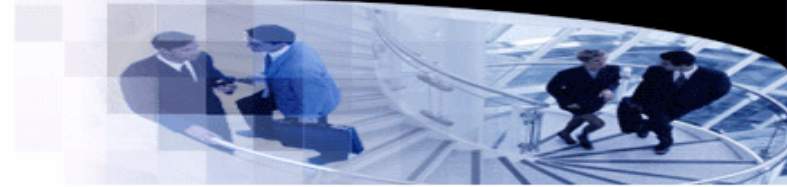
Live Variable Analysis



- **Applications**

- Register allocation: Assigning physical registers to variables.
- Basic idea:
 - At each statement where a variable v is defined, we assign a free register to v .
 - If there is no free register, then we choose a dead variable w at this location, and assign the physical register w to v .
 - What if there is no such w that is assigned with a register? We will explain it in the lecture of register allocation.

Live Variable Analysis



- Problem definition

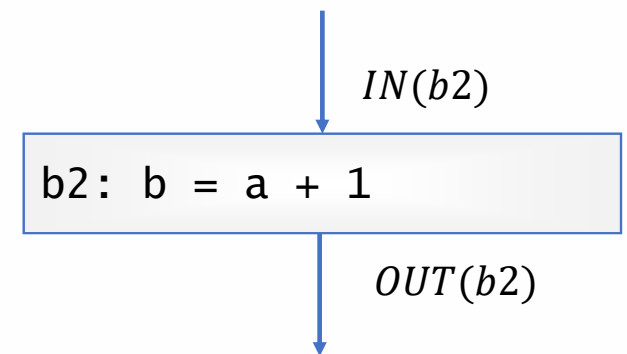
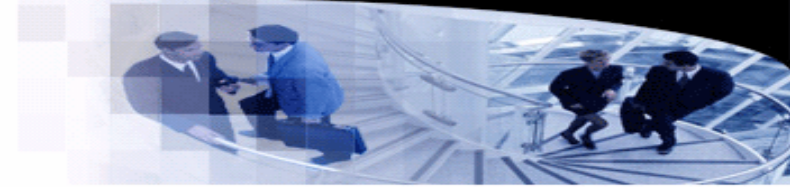
- Given a program location p , the data-flow value of p is the set of live variables.
- For the following program, live variable analysis calculates a function $L: \{b1, b2, b3, b4, b5, b6\} \rightarrow \{a, b, c\}$.
- If $L(b2) = \{a, c\}$, then a and c are live variables at the statement $b2$.

```
ENTRY:
b1: a = 0
b2: b = a + 1
b3: c = c + b
b4: a = b + 2
b5: br a, d6, d2
b6: return c
EXIT:
```

Live Variable Analysis

- Live variable equations

- What is the relation between $IN(b2)$ and $OUT(b2)$?
- If $IN(b2) = \{a, c\}$, then what is $OUT(b2)$?
- We know c is still alive after $b2$.
- We are not sure if a is still alive after $b2$.
- What about b ? We still do not know whether b will be used in the future.
- The forward transformation does not work in this case.
- Therefore, live variable analysis treats statements as backward transformation of live variables.



Live Variable Analysis

- Live variable equations

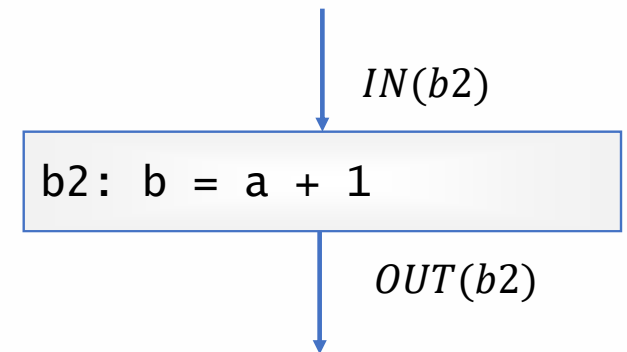
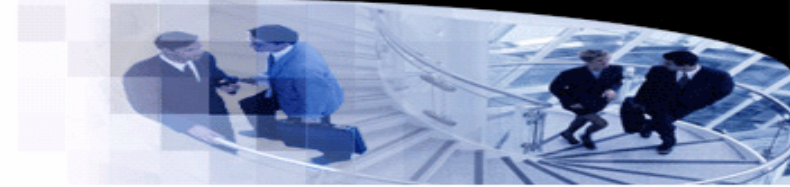
- Given a statement s , let $def_s = \{x \mid x \text{ is defined in } s.\}$.
- Let $use_s = \{x \mid x \text{ is used in } s.\}$.

- Example:

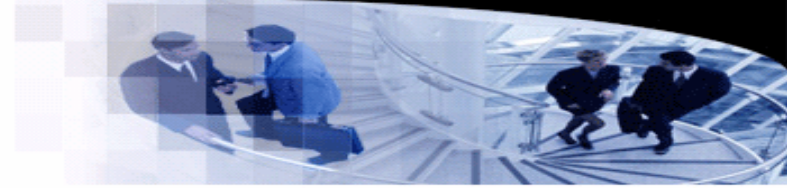
- $def_{b2} = \{b\}$.
- $use_{b2} = \{a\}$.

- Observation:

- If $OUT(b2) = \{b, c\}$, then what is $IN(b2)$?



Live Variable Analysis



- Live variable equations

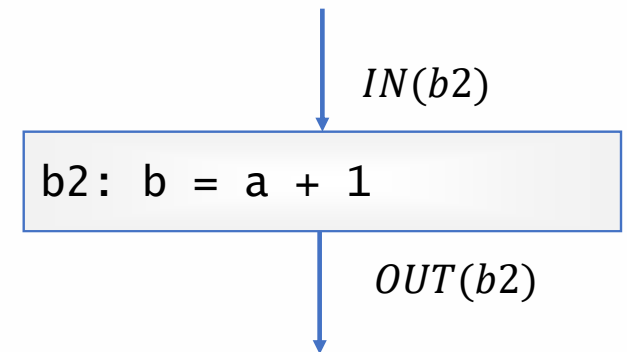
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- Let $use_s = \{x \mid x \text{ is used in } s.\}$.

- Example:

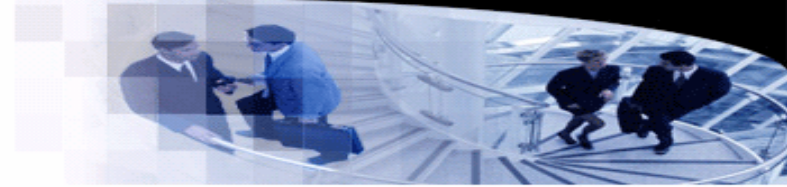
- $def_{b2} = \{b\}$.
- $use_{b2} = \{a\}$.

- Observation:

- If $OUT(b2) = \{b, c\}$, then what is $IN(b2)$?
- c is still alive before $b2$, because c is used somewhere in the future.



Live Variable Analysis



- Live variable equations

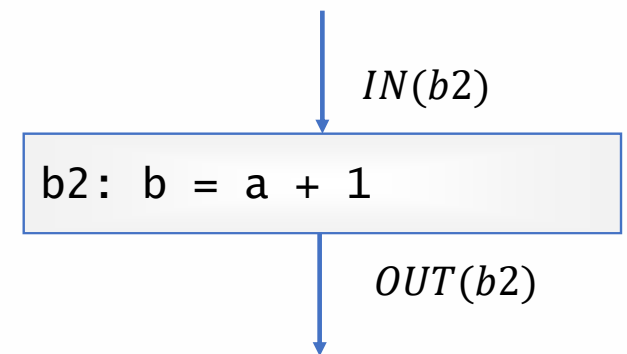
- Given a statement s , let $def_s = \{x \mid x \text{ is defined in } s.\}$.
- Let $use_s = \{x \mid x \text{ is used in } s.\}$.

- Example:

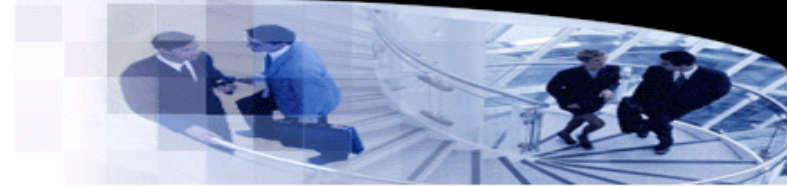
- $def_{b2} = \{b\}$.
- $use_{b2} = \{a\}$.

- Observation:

- If $OUT(b2) = \{b, c\}$, then what is $IN(b2)$?
- c is still alive before $b2$, because c is used somewhere in the future.
- a is alive before $b2$, because a is used in $b2$.



Live Variable Analysis



- Live variable equations

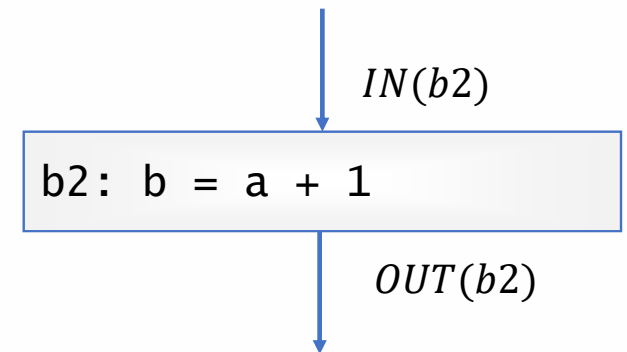
- Given a statement s , let $def_s = \{x \mid x \text{ is defined in } s.\}$.
- Let $use_s = \{x \mid x \text{ is used in } s.\}$.

- Example:

- $def_{b2} = \{b\}$.
- $use_{b2} = \{a\}$.

- Observation:

- If $OUT(b2) = \{b, c\}$, then what is $IN(b2)$?
- c is still alive before $b2$, because c is used somewhere in the future.
- a is alive before $b2$, because a is used in $b2$.
- However, b is dead before $b2$, because b is redefined at $b2$.



Live Variable Analysis

- Facts (data-flow relations) on statements

- $IN(s) = (OUT(s) - def_s) \cup use_s$

- Facts on basic blocks

- Let $B = s_1; s_2; \dots; s_n$.

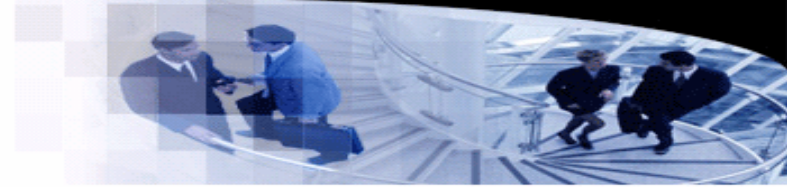
- $use_B = \bigcup_{s \in B} use_s$.

- $def_B = \bigcup_{s \in B} def_s$.

- $IN(B) = (OUT(B) - def_B) \cup use_B$

- Facts on control flows

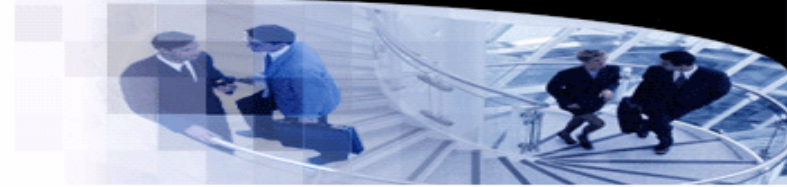
- $OUT(B) = \bigcup_{S \in succ(B)} IN(S)$.



Live Variable Analysis

- Algorithm

- $IN(EXIT) = \emptyset;$
- foreach basic block B do
- $IN(B) = \emptyset;$
- endfor
- while some $IN(B)$ changed do
- foreach basic block B do
- $OUT(B) = \bigcup_{S \in succ(B)} IN(S);$
- $IN(B) = (OUT(B) - def_B) \cup use_B;$
- endfor
- endwhile



Live Variable Analysis

- Running example

- Let us calculate *def* and *use* sets.

$use_{b1} = \{\}$
 $def_{b1} = \{a\}$

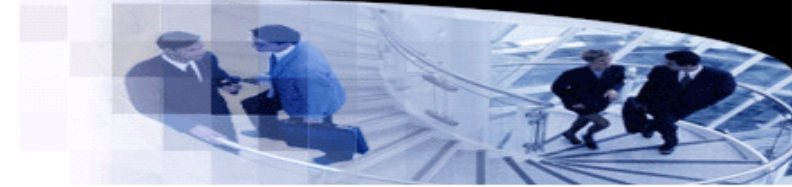
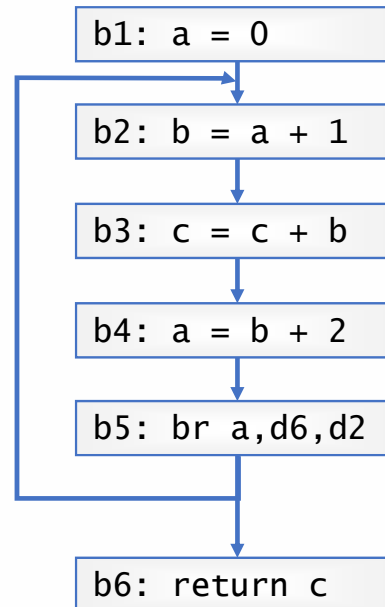
$use_{b2} = \{a\}$
 $def_{b2} = \{b\}$

$use_{b3} = \{c, b\}$
 $def_{b3} = \{c\}$

$use_{b4} = \{b\}$
 $def_{b4} = \{a\}$

$use_{b5} = \{a\}$
 $def_{b5} = \{\}$

$use_{b6} = \{c\}$
 $def_{b6} = \{\}$



Live Variable Analysis

- Running example

- Calculate live variables.

$use_{b1} = \{\}$
 $def_{b1} = \{a\}$

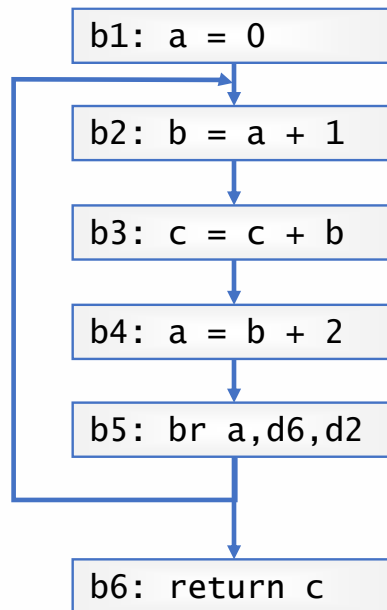
$use_{b2} = \{a\}$
 $def_{b2} = \{b\}$

$use_{b3} = \{c, b\}$
 $def_{b3} = \{c\}$

$use_{b4} = \{b\}$
 $def_{b4} = \{a\}$

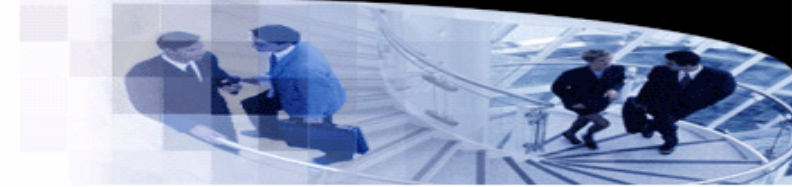
$use_{b5} = \{a\}$
 $def_{b5} = \{\}$

$use_{b6} = \{c\}$
 $def_{b6} = \{\}$



	OUT^0	IN^0	OUT^1	IN^1	OUT^2	IN^2	OUT^3	IN^3
b6	$\emptyset$	$\emptyset$						
b5	$\emptyset$	$\emptyset$						
b4	$\emptyset$	$\emptyset$						
b3	$\emptyset$	$\emptyset$						
b2	$\emptyset$	$\emptyset$						
b1	$\emptyset$	$\emptyset$						

Initialization.



Live Variable Analysis

• Running example

- Calculate live variables.

$use_{b1} = \{\}$
 $def_{b1} = \{a\}$

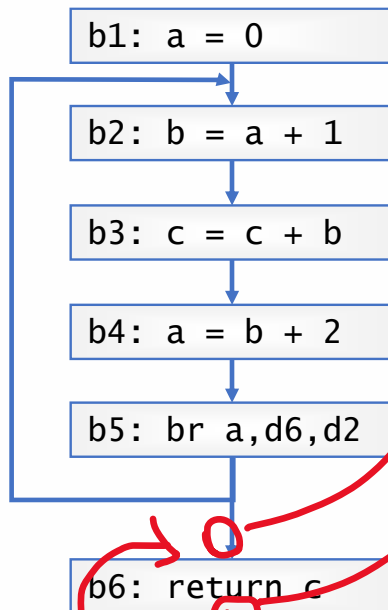
$use_{b2} = \{a\}$
 $def_{b2} = \{b\}$

$use_{b3} = \{c, b\}$
 $def_{b3} = \{c\}$

$use_{b4} = \{b\}$
 $def_{b4} = \{a\}$

$use_{b5} = \{a\}$
 $def_{b5} = \{\}$

$use_{b6} = \{c\}$
 $def_{b6} = \{\}$



	OUT^0	IN^0	OUT^1	IN^1	OUT^2	IN^2	OUT^3	IN^3
b6	$\emptyset$	$\emptyset$	$\emptyset$	$\{c\}$				
b5	$\emptyset$	$\emptyset$						
b4	$\emptyset$	$\emptyset$						
b3	$\emptyset$	$\emptyset$						
b2	$\emptyset$	$\emptyset$						
b1	$\emptyset$	$\emptyset$						

$$OUT^1(b6) = OUT^0(b6)$$

$$IN^1(b6) = OUT^1(b6) - def_{b6} \cup use_{b6} = \{c\}$$

Live Variable Analysis

• Running example

- Calculate live variables.

$use_{b1} = \{\}$
 $def_{b1} = \{a\}$

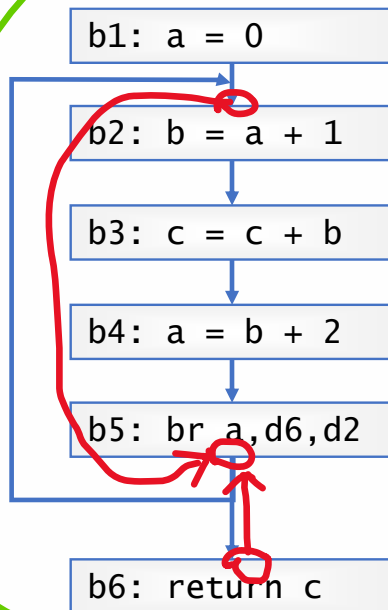
$use_{b2} = \{a\}$
 $def_{b2} = \{b\}$

$use_{b3} = \{c, b\}$
 $def_{b3} = \{c\}$

$use_{b4} = \{b\}$
 $def_{b4} = \{a\}$

$use_{b5} = \{a\}$
 $def_{b5} = \{\}$

$use_{b6} = \{c\}$
 $def_{b6} = \{\}$



	OUT^0	IN^0	OUT^1	IN^1	OUT^2	IN^2	OUT^3	IN^3
b6	$\emptyset$	$\emptyset$	$\emptyset$	$\{c\}$				
b5	$\emptyset$	$\emptyset$	$\{c\}$	$\{a, c\}$				
b4	$\emptyset$	$\emptyset$						
b3	$\emptyset$	$\emptyset$						
b2	$\emptyset$	$\emptyset$						
b1	$\emptyset$	$\emptyset$						

$$OUT^1(b5) = IN^0(b2) \cup IN^1(b6) = \{c\}$$

$$IN^1(b5) = OUT^1(b5) - def_{b5} \cup use_{b5} = \{a, c\}$$

Live Variable Analysis

• Running example

- Calculate live variables.

$use_{b1} = \{\}$
 $def_{b1} = \{a\}$

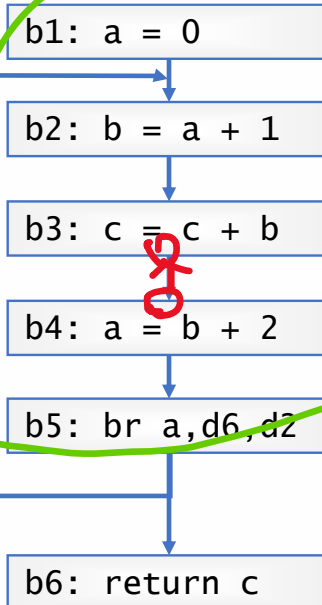
$use_{b2} = \{a\}$
 $def_{b2} = \{b\}$

$use_{b3} = \{c, b\}$
 $def_{b3} = \{c\}$

$use_{b4} = \{b\}$
 $def_{b4} = \{a\}$

$use_{b5} = \{a\}$
 $def_{b5} = \{\}$

$use_{b6} = \{c\}$
 $def_{b6} = \{\}$



	OUT^0	IN^0	OUT^1	IN^1	OUT^2	IN^2	OUT^3	IN^3
b6	$\emptyset$	$\emptyset$	$\emptyset$	$\{c\}$				
b5	$\emptyset$	$\emptyset$	$\{c\}$	$\{a, c\}$				
b4	$\emptyset$	$\emptyset$	$\{a, c\}$	$\{b, c\}$				
b3	$\emptyset$	$\emptyset$						
b2	$\emptyset$	$\emptyset$						
b1	$\emptyset$	$\emptyset$						

$$OUT^1(b4) = IN^1(b5) = \{a, c\}$$

$$IN^1(b4) = OUT^1(b4) - def_{b4} \cup use_{b4}$$

$$= \{a, c\} - \{a\} \cup \{b\} = \{b, c\}$$

Live Variable Analysis

• Running example

- Calculate live variables.

$use_{b1} = \{\}$
 $def_{b1} = \{a\}$

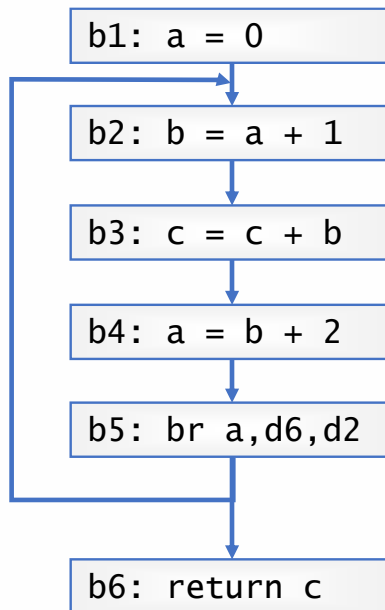
$use_{b2} = \{a\}$
 $def_{b2} = \{b\}$

$use_{b3} = \{c, b\}$
 $def_{b3} = \{c\}$

$use_{b4} = \{b\}$
 $def_{b4} = \{a\}$

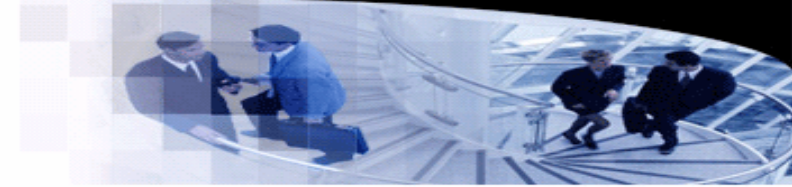
$use_{b5} = \{a\}$
 $def_{b5} = \{\}$

$use_{b6} = \{c\}$
 $def_{b6} = \{\}$

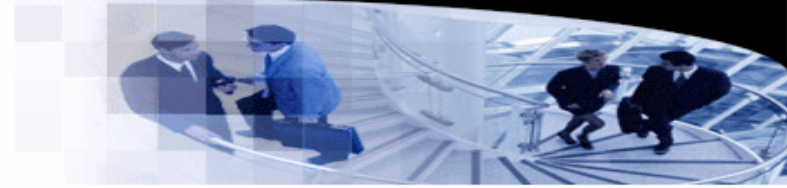


	OUT^0	IN^0	OUT^1	IN^1	OUT^2	IN^2	OUT^3	IN^3
b6	$\emptyset$	$\emptyset$	$\emptyset$	$\{c\}$	$\emptyset$	$\{c\}$	$\emptyset$	$\{c\}$
b5	$\emptyset$	$\emptyset$	$\{c\}$	$\{a, c\}$	$\{a, c\}$	$\{a, c\}$	$\{a, c\}$	$\{a, c\}$
b4	$\emptyset$	$\emptyset$	$\{a, c\}$	$\{b, c\}$	$\{a, c\}$	$\{b, c\}$	$\{a, c\}$	$\{b, c\}$
b3	$\emptyset$	$\emptyset$	$\{b, c\}$	$\{b, c\}$	$\{b, c\}$	$\{b, c\}$	$\{b, c\}$	$\{b, c\}$
b2	$\emptyset$	$\emptyset$	$\{b, c\}$	$\{a, b\}$	$\{b, c\}$	$\{a, b\}$	$\{b, c\}$	$\{a, b\}$
b1	$\emptyset$	$\emptyset$	$\{a, c\}$	$\{c\}$	$\{a, c\}$	$\{c\}$	$\{a, c\}$	$\{c\}$

Now let us check the results of live variables by IN^3 .



Live Variable Analysis



- **Interference**

- Two variables interfere if they are live at the same location.
- Interfering variables cannot share the same physical register.
- Facts by the result of live variable analysis:
 - a and c cannot share a register.
 - b and c cannot share a register.
 - a and b cannot share a register.
 - Therefore, they have to be assigned to three different registers.
- Register allocation via graph coloring.

	OUT^3	IN^3
b6	$\emptyset$	{c}
b5	{a,c}	{a,c}
b4	{a,c}	{b,c}
b3	{b,c}	{b,c}
b2	{b,c}	{a,b}
b1	{a,c}	{c}